

Inferential Statistics: Knowledge Score vs. Marital Status

Independent Two-Sample T-Test

This report examines whether there is a statistically significant difference in clinical knowledge between Married and Single thalassemia carriers.

Descriptive Statistics

Marital Status	N	Mean (V3 Score)	Std Dev (V3 Score)	Mean (Weighted V3)	Std Dev (Weighted)
Married	70	9.77	7.99	1.067	4.381
Single	131	12.17	8.14	2.456	4.604

Hypothesis Testing (Welch's T-Test)

We use Welch's T-Test (assuming unequal variances) for robust results.

1. Analysis using the mathematically rigorous Weighted V3 Knowledge Score:

- **T-Statistic:** -2.104
- **P-Value:** 0.0371
- **Conclusion: Statistically Significant ($p < 0.05$).** There is a meaningful difference in knowledge between married and single carriers.

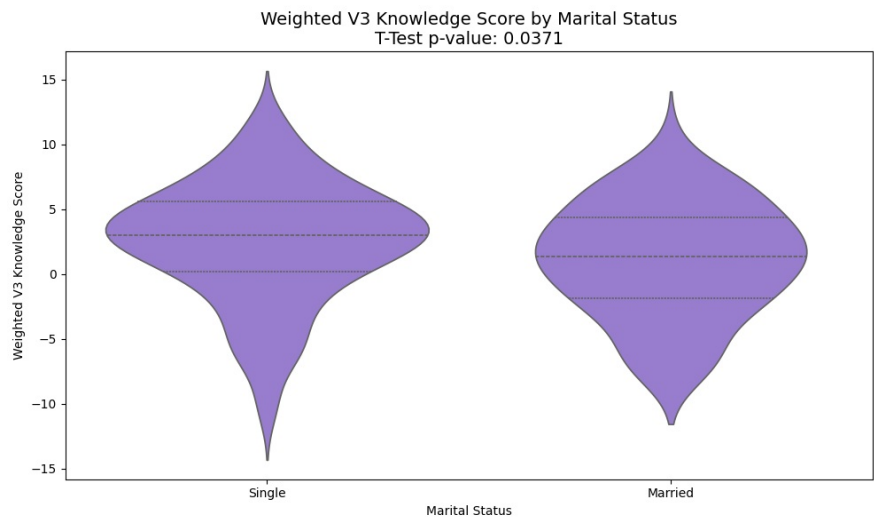
2. Analysis using the standard V3 Knowledge Score:

- **T-Statistic:** -2.014
- **P-Value:** 0.0459
- **Conclusion: Statistically Significant ($p < 0.05$).**

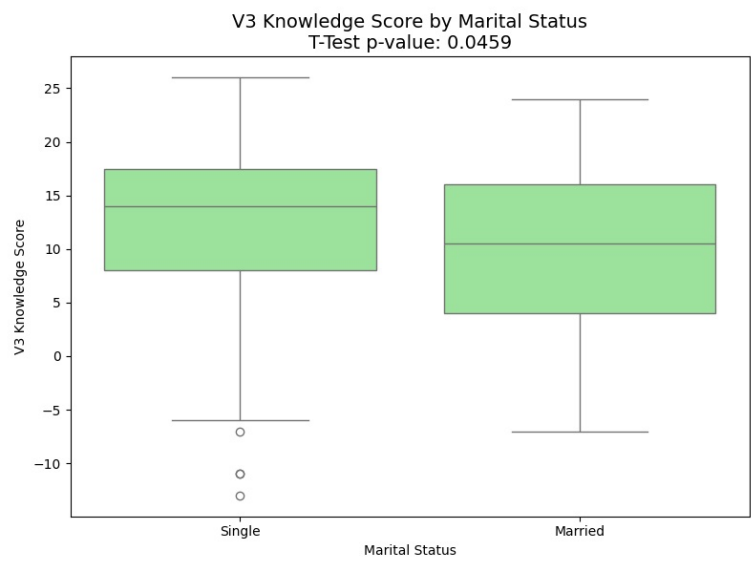
Visualizations

Weighted V3 Score Violin Plot

The violin plot demonstrates the distribution density and quartiles for both groups using the rigorously weighted metric.



Standard V3 Score Box Plot



Box Plot